

PORTOFOLIO

STUDI KASUS

Pola Operasi Kilang Minyak Menggunakan Algoritma K-Means

1

PEMANGGILAN DATA

```
Drive already mounted at /content/drive; to attempt to
percentage yield gravity vapour pressure \
0          6.9      38.4          6.1
1         14.4      40.3          4.8
2          7.4      40.0          6.1
3          8.5      31.8          0.2
4          8.0      40.8          3.5

ten percent distillation point fraction end point
0                220                235
1                231                307
2                217                212
3                316                365
4                210                218
```

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DATA PRE-PROCESSING

```
Missing values before handling:
percentage yield      0
gravity              0
vapour pressure      0
ten percent distillation point  0
fraction end point   0
cluster              0
dtype: int64

DataFrame after scaling:
percentage yield gravity vapour pressure ten percent distillation point fraction end point cluster
0          -1.209013 -0.153245      0.744114          -0.581865          -1.414178      1
1          -0.498351  0.189303      0.239959          -0.284167          -0.365492      1
2          -1.161635  0.135216      0.744114          -0.663056          -1.749175      1
3          -1.057405 -1.343147     -1.543976           2.016231           0.479282      0
4          -1.104782  0.279447     -0.264197          -0.852500          -1.661784      1
```

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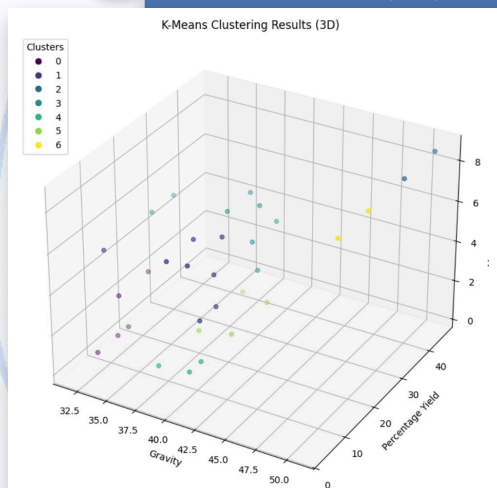
K-means CLUSTERING

Original DataFrame with cluster labels:

	percentage yield	gravity	vapour pressure	ten percent distillation point	fraction end point	cluster
0	6.9	38.4	6.1	220	235	1
1	14.4	40.3	4.8	231	307	1
2	7.4	40.0	6.1	217	212	1
3	8.5	31.8	0.2	316	365	0
4	8.0	40.8	3.5	210	218	1

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VISUALISASI K-means CLUSTERING (3D)



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VISUALISASI INTERACTIVE K-means CLUSTERING

